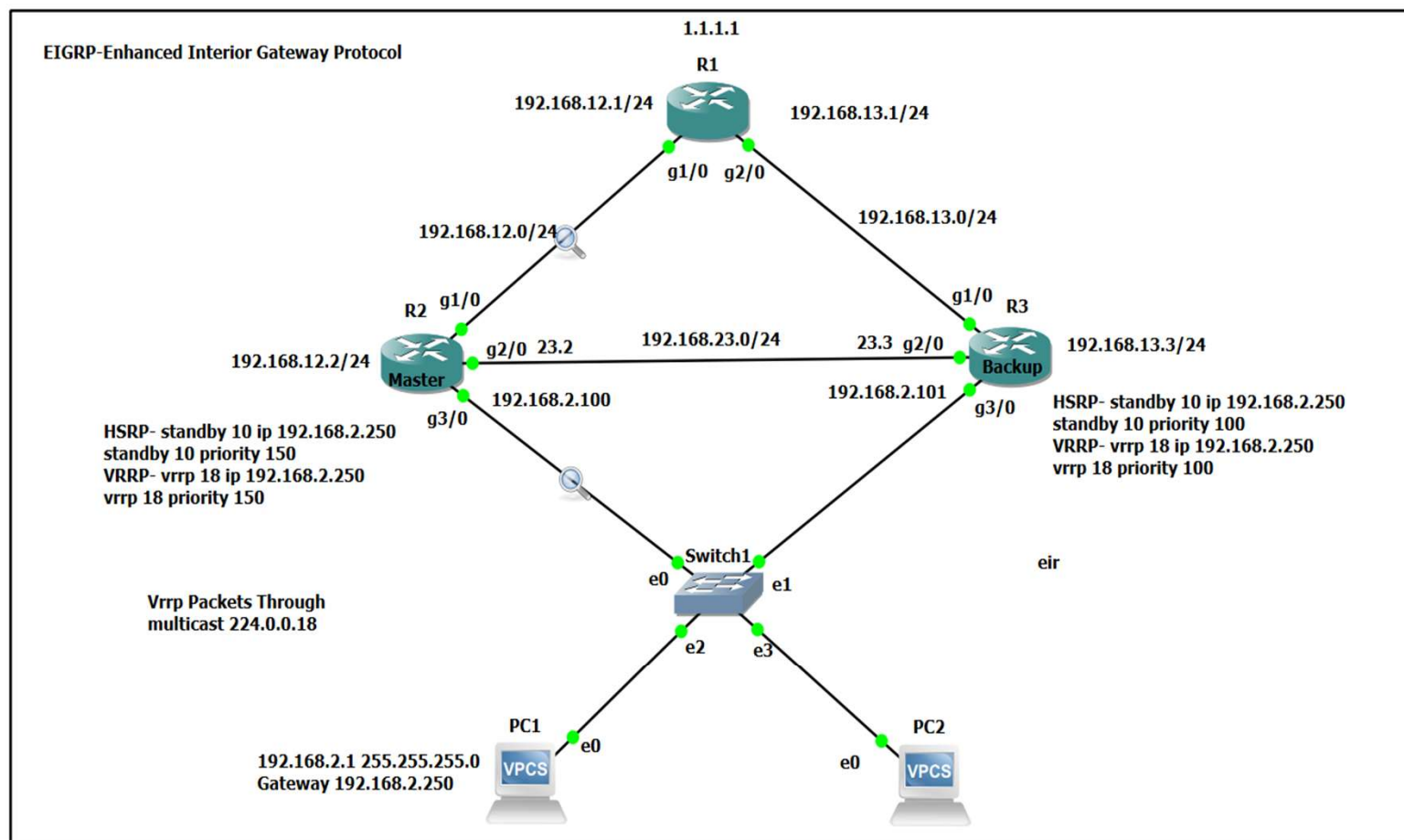




```
R1 R2 R3 +
R2(config)#do show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
+ - replicated route, % - next hop override

Gateway of last resort is not set

1.0.0.0/32 is subnetted, 1 subnets
  1.1.1.1 [90/130816] via 192.168.12.1, 00:00:44, GigabitEthernet1/0
192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.2.0/24 is directly connected, GigabitEthernet3/0
L    192.168.2.100/32 is directly connected, GigabitEthernet3/0
192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.12.0/24 is directly connected, GigabitEthernet1/0
L    192.168.12.2/32 is directly connected, GigabitEthernet1/0
D    192.168.13.0/24 [90/3072] via 192.168.12.1, 00:00:44, GigabitEthernet1/0
192.168.23.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.23.0/24 is directly connected, GigabitEthernet2/0
L    192.168.23.2/32 is directly connected, GigabitEthernet2/0
R2(config)#
R2(config)#
*Sep 16 12:07:18.111: %DUAL-5-NBRCHANGE: EIGRP-IPv4 90: Neighbor 192.168.23.3 (G
igabitEthernet2/0) is up: new adjacency
R2(config)#
*Sep 16 12:07:33.655: %DUAL-5-NBRCHANGE: EIGRP-IPv4 90: Neighbor 192.168.2.101 (
GigabitEthernet3/0) is up: new adjacency
R2(config)#int g3/0
R2(config-if)#vrrp 18 ip 192.168.2.250
R2(config-if)#v
*Sep 16 12:14:52.051: %VRRP-6-STATECHANGE: Gi3/0 Grp 18 state Init -> Backup
*Sep 16 12:14:52.063: %VRRP-6-STATECHANGE: Gi3/0 Grp 18 state Backup -> Init
*Sep 16 12:14:52.063: %VRRP-6-STATECHANGE: Gi3/0 Grp 18 state Init -> Backup
R2(config-if)#vrrp 18 priority 150
R2(config-if)#
*Sep 16 12:15:06.807: %VRRP-6-STATECHANGE: Gi3/0 Grp 18 state Backup -> Master
R2(config-if)#exit
R2(config)#do show vrrp br
Interface      Grp Pri Time  Own Pre State  Master addr  Group addr
Gi3/0          18 150 3414      Y Master 192.168.2.100 192.168.2.250
R2(config)#
```

VRRP OF EIGRP

```
R1 R2 R3 +
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
+ - replicated route, % - next hop override

Gateway of last resort is not set

1.0.0.0/32 is subnetted, 1 subnets
D    1.1.1.1 [90/130816] via 192.168.13.1, 00:00:15, GigabitEthernet1/0
192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.2.0/24 is directly connected, GigabitEthernet3/0
L    192.168.2.101/32 is directly connected, GigabitEthernet3/0
D    192.168.12.0/24 [90/3072] via 192.168.23.2, 00:00:15, GigabitEthernet2/0
[90/3072] via 192.168.13.1, 00:00:15, GigabitEthernet1/0
[90/3072] via 192.168.2.100, 00:00:15, GigabitEthernet3/0
192.168.13.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.13.0/24 is directly connected, GigabitEthernet1/0
L    192.168.13.3/32 is directly connected, GigabitEthernet1/0
192.168.23.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.23.0/24 is directly connected, GigabitEthernet2/0
L    192.168.23.3/32 is directly connected, GigabitEthernet2/0
R3(config)#
R3(config)#vrrp 18 ip 192.168.2.250
^
% Invalid input detected at '^' marker.

R3(config)#int g3/0
R3(config-if)#vrrp 18 ip 192.168.2.250
R3(config-if)#v
*Sep 16 12:12:28.379: %VRRP-6-STATECHANGE: Gi3/0 Grp 18 state Init -> Backup
*Sep 16 12:12:28.387: %VRRP-6-STATECHANGE: Gi3/0 Grp 18 state Backup -> Init
*Sep 16 12:12:28.387: %VRRP-6-STATECHANGE: Gi3/0 Grp 18 state Init -> Backup
R3(config-if)#vrrp
*Sep 16 12:12:31.999: %VRRP-6-STATECHANGE: Gi3/0 Grp 18 state Backup -> Master
R3(config-if)#vrrp 18 priority 100
R3(config-if)#
*Sep 16 12:15:06.831: %VRRP-6-STATECHANGE: Gi3/0 Grp 18 state Master -> Backup
R3(config-if)#exit
R3(config)#do show vrrp br
Interface      Grp Pri Time  Own Pre State  Master addr  Group addr
Gi3/0          18 100 3609      Y Backup 192.168.2.100 192.168.2.250
R3(config)#
```

```

R1 R2 R3 PC1
*Sep 16 13:15:08.987: %LINK-5-CHANGED: Interface GigabitEthernet3/0, changed state to administratively down
*Sep 16 13:15:09.987: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet3/0, changed state to down
R2(config-if)#no shutdown
^
% Invalid input detected at '^' marker.

R2(config-if)#no shutdown
R2(config-if)#
*Sep 16 13:16:09.147: %LINK-3-UPDOWN: Interface GigabitEthernet3/0, changed state to up
*Sep 16 13:16:10.147: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet3/0, changed state to up
R2(config-if)#
*Sep 16 13:16:11.343: %DUAL-5-NBRCHANGE: EIGRP-IPv4 90: Neighbor 192.168.2.101 (GigabitEthernet3/0) is up: new adjacency
R2(config-if)#
*Sep 16 13:16:29.019: %HSRP-5-STATECHANGE: GigabitEthernet3/0 Grp 10 state Speak -> Standby
R2(config-if)#exit
R2(config)#do show standby br
P indicates configured to preempt.
|
Interface Grp Pri P State Active Standby Virtual IP
Gi3/0 10 150 Standby 192.168.2.101 local 192.168.2.250
R2(config)#
*Sep 16 13:17:54.547: %HSRP-5-STATECHANGE: GigabitEthernet3/0 Grp 10 state Standby -> Active
R2(config)#
*Sep 16 13:18:04.983: %DUAL-5-NBRCHANGE: EIGRP-IPv4 90: Neighbor 192.168.2.101 (GigabitEthernet3/0) is down: holding time expired
R2(config)#int g3/0
R2(config-if)#shutdown
R2(config-if)#
*Sep 16 13:18:34.839: %HSRP-5-STATECHANGE: GigabitEthernet3/0 Grp 10 state Active -> Init
R2(config-if)#
*Sep 16 13:18:36.831: %LINK-5-CHANGED: Interface GigabitEthernet3/0, changed state to administratively down
*Sep 16 13:18:37.831: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet3/0, changed state to down
R2(config-if)#exit
R2(config)#do show standby br
P indicates configured to preempt.
|
Interface Grp Pri P State Active Standby Virtual IP
Gi3/0 10 150 Init unknown unknown 192.168.2.250
R2(config)#

```

## HSRP OF EIGRP

```

R1 R2 R3 PC1
GigabitEthernet3/0) is up: new adjacency
R3(config)#
*Sep 16 13:09:31.655: %DUAL-5-NBRCHANGE: EIGRP-IPv4 90: Neighbor 192.168.2.100 (GigabitEthernet3/0) is down: holding time expired
R3(config)#do show standby br
P indicates configured to preempt.
|
Interface Grp Pri P State Active Standby Virtual IP
Gi3/0 10 100 Active local unknown 192.168.2.250
R3(config)#
*Sep 16 13:12:24.331: %DUAL-5-NBRCHANGE: EIGRP-IPv4 90: Neighbor 192.168.2.100 (GigabitEthernet3/0) is up: new adjacency
R3(config)#
*Sep 16 13:15:08.643: %DUAL-5-NBRCHANGE: EIGRP-IPv4 90: Neighbor 192.168.2.100 (GigabitEthernet3/0) is down: holding time expired
R3(config)#
*Sep 16 13:16:01.047: %DUAL-5-NBRCHANGE: EIGRP-IPv4 90: Neighbor 192.168.2.100 (GigabitEthernet3/0) is up: new adjacency
R3(config)#do show standby br
P indicates configured to preempt.
|
Interface Grp Pri P State Active Standby Virtual IP
Gi3/0 10 100 Active local 192.168.2.100 192.168.2.250
R3(config)#int g3/0
R3(config-if)#shutdown
R3(config-if)#
*Sep 16 13:17:44.295: %HSRP-5-STATECHANGE: GigabitEthernet3/0 Grp 10 state Active -> Init
*Sep 16 13:17:44.323: %DUAL-5-NBRCHANGE: EIGRP-IPv4 90: Neighbor 192.168.2.100 (GigabitEthernet3/0) is down: interface down
R3(config-if)#
*Sep 16 13:17:46.287: %LINK-5-CHANGED: Interface GigabitEthernet3/0, changed state to administratively down
*Sep 16 13:17:47.287: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet3/0, changed state to down
R3(config-if)#no shutdown
R3(config-if)#
*Sep 16 13:18:35.771: %LINK-3-UPDOWN: Interface GigabitEthernet3/0, changed state to up
*Sep 16 13:18:36.771: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet3/0, changed state to up
R3(config-if)#
*Sep 16 13:18:56.379: %HSRP-5-STATECHANGE: GigabitEthernet3/0 Grp 10 state Standby -> Active
R3(config-if)#exit
R3(config)#do show standby br
P indicates configured to preempt.
|
Interface Grp Pri P State Active Standby Virtual IP
Gi3/0 10 100 Active local unknown 192.168.2.250
R3(config)#

```

R1

R2

R3

PC1

84 bytes from 1.1.1.1 icmp\_seq=36 ttl=254 time=45.585 ms  
84 bytes from 1.1.1.1 icmp\_seq=37 ttl=254 time=45.679 ms  
84 bytes from 1.1.1.1 icmp\_seq=38 ttl=254 time=49.187 ms  
84 bytes from 1.1.1.1 icmp\_seq=39 ttl=254 time=49.304 ms  
84 bytes from 1.1.1.1 icmp\_seq=40 ttl=254 time=48.378 ms  
84 bytes from 1.1.1.1 icmp\_seq=41 ttl=254 time=47.740 ms  
84 bytes from 1.1.1.1 icmp\_seq=42 ttl=254 time=48.305 ms  
84 bytes from 1.1.1.1 icmp\_seq=43 ttl=254 time=47.784 ms  
84 bytes from 1.1.1.1 icmp\_seq=44 ttl=254 time=47.976 ms  
84 bytes from 1.1.1.1 icmp\_seq=45 ttl=254 time=47.182 ms  
84 bytes from 1.1.1.1 icmp\_seq=46 ttl=254 time=48.734 ms  
84 bytes from 1.1.1.1 icmp\_seq=47 ttl=254 time=47.350 ms  
84 bytes from 1.1.1.1 icmp\_seq=48 ttl=254 time=48.883 ms  
84 bytes from 1.1.1.1 icmp\_seq=49 ttl=254 time=46.674 ms  
84 bytes from 1.1.1.1 icmp\_seq=50 ttl=254 time=48.262 ms  
84 bytes from 1.1.1.1 icmp\_seq=51 ttl=254 time=48.077 ms  
84 bytes from 1.1.1.1 icmp\_seq=52 ttl=254 time=46.956 ms  
1.1.1.1 icmp\_seq=53 timeout  
1.1.1.1 icmp\_seq=54 timeout  
1.1.1.1 icmp\_seq=55 timeout  
1.1.1.1 icmp\_seq=56 timeout  
1.1.1.1 icmp\_seq=57 timeout  
1.1.1.1 icmp\_seq=58 timeout  
1.1.1.1 icmp\_seq=59 timeout  
1.1.1.1 icmp\_seq=60 timeout  
1.1.1.1 icmp\_seq=61 timeout  
1.1.1.1 icmp\_seq=62 timeout  
1.1.1.1 icmp\_seq=63 timeout  
1.1.1.1 icmp\_seq=64 timeout  
1.1.1.1 icmp\_seq=65 timeout  
1.1.1.1 icmp\_seq=66 timeout  
1.1.1.1 icmp\_seq=67 timeout  
1.1.1.1 icmp\_seq=68 timeout  
84 bytes from 1.1.1.1 icmp\_seq=69 ttl=254 time=78.255 ms  
84 bytes from 1.1.1.1 icmp\_seq=70 ttl=254 time=47.656 ms  
84 bytes from 1.1.1.1 icmp\_seq=71 ttl=254 time=45.551 ms  
84 bytes from 1.1.1.1 icmp\_seq=72 ttl=254 time=49.255 ms  
84 bytes from 1.1.1.1 icmp\_seq=73 ttl=254 time=48.702 ms  
84 bytes from 1.1.1.1 icmp\_seq=74 ttl=254 time=47.309 ms  
84 bytes from 1.1.1.1 icmp\_seq=75 ttl=254 time=47.693 ms  
84 bytes from 1.1.1.1 icmp\_seq=76 ttl=254 time=46.272 ms  
84 bytes from 1.1.1.1 icmp\_seq=77 ttl=254 time=48.830 ms  
  
PC1>  
PC1>  
PC1>

HSRP Active Router Failure : ICMP Packet Loss During Failover and Network Convergence